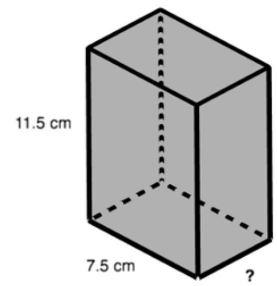
Grade 6 Accelerated Name____________________
Unit 8 Date_____________________
Lesson 3 Homework Period____________________

A rectangular prism is shown with measurements in centimeters (cm).

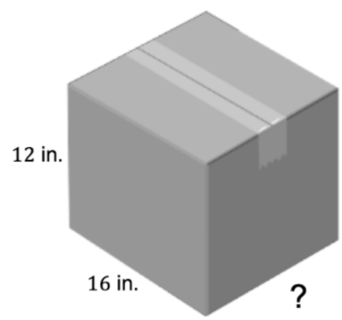
1. What is the area of the front face of this prism? Include units.



2. If the volume of the box is 345 cubic centimeters (cm^3), what is the value of the missing dimension of the box?

Stefani is packing moving boxes. The boxes have a length of 16 inches (in.) and a height of 12 in., but she cannot recall the width of the box.

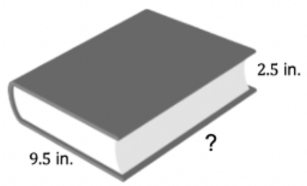
3. Determine the area of the front face of the box. Include units.



4. If the volume of the box is $2{,}304$ cubic inches (in^3), what is the width of the box? Include units.

Jeremiah was trying to find the missing dimension of the journal shown. He made a mistake when solving for the width of the box, but he cannot find it. His work is shown.

Volume = 285 cubic in.



Jeremiah's work	
$\begin{array}{r} 9.5 \\ \times\ \ 2.5 \\ \hline 455 \\ +\,1900 \\ \hline 2355 \rightarrow 23.55\ \text{in}^2 \end{array}$	$285 \div 23.55 = 12.1$ $w = 12.1$ in.

5. Circle the mistake.

6. What should the width of the box be? Include units.

7. Write a note to Jeremiah on how to correct his error(s).

8. A box of crayons has a length of 1.5 inches (in.) and a height of 4 in. The box has a volume of 16.8 cubic inches (in^3). What is the width of the box? Include units.

9. Josh bought a sandbox. He knows that the length is 3.5 feet (ft) and the width is 2.5 ft., but he can't remember the appropriate height. The volume needs to be 10.5 cubic feet (ft^3). What is the height of the sandbox? Include units.

10. Cara is measuring her suitcase before her trip. The case has a width of 8.75 inches (in.) and a height of 14 in. The volume of the case is $2{,}756.25$ cubic inches (in^3). Determine the length of the suitcase. Include units.